

Semantics-Aware Control Barrier Functions on Piecewise Polynomial Boundaries via Bernstein Coefficient Constraints

Abstract—Safe navigation in real-world environments requires robots to reason not only about obstacle geometry, but also the distinct safety implications across different parts of an obstacle. Current safety methods often employ a uniform clearance for an entire obstacle. This makes them too cautious in low-risk areas and not protective enough in hazardous regions. In this paper, we present a semantics-aware control barrier function (CBF) method that enables spatially varying safety margins along irregular obstacle boundaries. This enables robots to maintain larger clearances near hazardous regions while preserving navigable space around safer regions compared with uniformly applying the largest margin. To enforce these safety constraints over continuous obstacle boundaries, we derive Bernstein coefficient constraints that provide a finite sufficient condition for the CBF inequality over the entire fitted boundary. We evaluate the proposed method in simulated navigation scenarios that include crossing pedestrians and cluttered household settings. The results show that considering how risk changes in different regions can encourage the robot to avoid dangerous parts of obstacles.

I. INTRODUCTION

Semantics-aware safety has gained increasing attention in robotics because collision avoidance heavily depends on the nature of the obstacle [1]. For example, a robot brushing against the side of a sofa and a robot bumping into the front of a person carrying a cup of hot water should not be governed by the same clearance. Traditional collision avoidance methods, however, encode safety purely through geometric distance, such as Control Barrier Functions (CBFs) built on signed distance or inflated ellipses [2], [3], where the clearance is dictated by geometry alone and cannot reflect that some parts of an obstacle are more dangerous than others. With modern perception, robots can reason based on semantic risk representations mapped from object categories and scene contexts. This makes it possible to encode the severity of risk within safety reasoning frameworks [4]–[6].

Recently work assign safety margins based on semantic risks [7], [8]. In [7], semantic labels are grouped into risk classes, and each class inflates the obstacle by its own margin inside a signed distance field (SDF). By assigning different local risks on the boundary, the robot can avoid unnecessarily conservative behavior near safe regions, which is beneficial in dense environments. However, the risk in this method is uniformly attached to the object, whereas real-world environments often include obstacles that are *risk-heterogeneous*, meaning the danger of a real obstacle is rarely uniform along its boundary. For example, the risk of colliding with a vehicle is usually much higher at the front and rear of the vehicle than at the sides, due to braking distances and the uncertainty of longitudinal movement [9]. A safety margin

should thus vary continuously along an irregular boundary, which a single per-object inflation cannot express.

In this context, researchers have developed semantic risk-aware margins that adjust distance based on obstacle segments [8]. In addition, sampling-based distance functions have been proposed to handle obstacles with nonconvex shapes [10]. On the other hand, Bernstein polynomial approximations have been used to construct geometry-aware CBFs from a signed-distance approximation [11]. This considers the shape of the obstacle, but not how risk varies along its boundary, and relies on a signed-distance-field approximation, which can overestimate or underestimate the true distance by varying amounts along the boundary. A recent polytopic approach handles exact signed distances with nonsmooth CBFs [12]. However, these methods do not provide a unified representation that captures both irregular geometry and spatially varying risk while still allowing safety to be certified over the full boundary using only a finite set of verifiable constraints.

To address this challenge, we introduce a continuously differentiable unified obstacle representation that incorporates irregular boundaries and local semantic risks amenable to provable control enforcement. Specifically, we jointly parameterize boundary geometry and semantic risk using periodic cubic B-splines, such that each location along the fitted boundary is directly associated with its local semantic risk under the shared continuous parameterization. This enables safety conditions to be formulated continuously over the obstacle boundary accounting for both geometry and semantics without additional queries. Then we exploit the piecewise polynomial structure of such representations to formulate resulting boundary-wise CBF conditions and finite Bernstein coefficient constraints that are sufficient to enforce them over the entire fitted boundary. This allows for transforming continuously varying geometric and semantic information into a finite set of differentiable and enforceable constraints compatible for CBF-based safety filtering.

Our **contributions** are summarized below.

- We introduce a Semantics-Aware Control Barrier Function (Semantics-Aware CBF) using a unified B-spline representation of irregular boundary geometry and spatially varying semantic risk fitted from boundary samples.
- We establish a finite set of Bernstein coefficient constraints that enforce the CBF condition over the entire fitted boundary for both physical and semantic safety.
- We provide theoretical proof of the system safety and simulation results to validate the performance and efficiency of our method.

II. PRELIMINARIES

Consider a robot with state $\mathbf{x} \in \mathbb{R}^n$ and dynamics

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}, \quad (1)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times q}$ are locally Lipschitz and $\mathbf{u} \in \mathcal{U} \subset \mathbb{R}^q$ with $\mathcal{U}$ compact and convex.

A. Control Barrier Functions

For a continuously differentiable function $h : \mathbb{R}^n \rightarrow \mathbb{R}$, the safe set is defined as $\mathcal{H} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}$.

Lemma 1 (Forward invariance under a control barrier function [13]). *Suppose that $\nabla h \neq \mathbf{0}$ on the boundary $\partial\mathcal{H}$ and an extended class- $\mathcal{K}$ function α exists that satisfies $\sup_{\mathbf{u} \in \mathcal{U}} \dot{h} \geq -\alpha(h)$ on an open set containing $\mathcal{H}$. Then, any locally Lipschitz feedback controller $\mathbf{u} = \mathbf{k}(\mathbf{x}) \in \mathcal{U}$ that satisfies $\dot{h} + \alpha(h(\mathbf{x})) \geq 0$ on that set, where $\dot{h} = \nabla_{\mathbf{x}} h^\top (f(\mathbf{x}) + g(\mathbf{x})\mathbf{k}(\mathbf{x}))$, renders the safe set $\mathcal{H}$ forward invariant. In other words, if the system starts within $\mathcal{H}$, it remains in $\mathcal{H}$ as long as the solution persists.*

In this paper, we adopt $\alpha(h) = \gamma h$ with $\gamma > 0$. We further require the Control Barrier Function (CBF) to have a relative degree one with respect to Eq. (1), i.e., $\nabla h^\top g(\mathbf{x}) \neq 0$, so that the control input appears in $\dot{h}$.

B. Problem Statement

We consider a robot with dynamics given by Eq. (1) as it moves in a planar workspace with obstacles defined by $\mathcal{O} = \bigcup_{l=1}^L \mathcal{O}_l$. Each obstacle $\mathcal{O}_l$ is compact and connected. The semantic risk associated with it can change at different points along its boundary, $\partial\mathcal{O}_l$. We use $\tau \in \mathbb{S}^1$ to parameterize points along each obstacle boundary.

Assumption 1. *For each $\mathcal{O}_l$, we assume there is a closed, continuous boundary curve $\mathbf{p} : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ and a semantic risk $s : \mathbb{S}^1 \rightarrow \mathbb{R}$. The risk ranges from -1 to 1 , where $s = -1$ means highest risk and $s = 1$ means lowest risk. These are assumed to be fitted offline using denoised point-cloud data, computer-aided-design models, or annotated maps [14]. The fitted boundary is checked against the known boundary. The fitted region must be a conservative outer approximation. This means that $\mathcal{O}_l$ must lie in the interior of $\Omega(\mathbf{p})$, where $\Omega(\mathbf{p})$ is the region enclosed by $\mathbf{p}$. This condition is certified offline. Online perception and the correctness of the semantic labels are outside the scope of this paper.*

This representation is best suited to obstacles with rigid geometry, like furniture. A fitted enclosure can be reused under rigid motion (see Sec. III-C) or semantic relabeling, such as relocating the cup shown in Fig. 4.

Problem 1 (Semantics-aware safety filter). *Given the offline-fitted boundary and risk field $(\mathbf{p}, s)$, design a safety filter that outputs a control input $\mathbf{u}^* \in \mathcal{U}$ such that (i) the robot maintains a greater distance from high-risk regions of the boundary and a smaller distance in low-risk regions, and (ii) the safety margin provided by the risk field guarantees collision avoidance with the true obstacle $\mathcal{O}_l$.*

The safety filter finds a control input close to the nominal command $\mathbf{u}^{\text{nom}}(\mathbf{x}) \in \mathcal{U}$ that satisfies the CBF constraint for all boundary parameters of the obstacle to ensure both physical and semantic safety. Here, $\mathcal{T}$ gathers these parameters and assigns one circle to each obstacle. The term $\mathbf{v}_{\text{obs},l}(\tau)$ is the boundary point velocity, which is zero when the obstacle is not moving. We map the semantic risk $s(\tau)$ to the margin $m(s(\tau))$ (see Sec. III-B). This margin is used to construct the barrier h (see Sec. III-B). We can then express Problem 1 as follows:

$$\begin{aligned} \mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathcal{U}} \quad & \|\mathbf{u} - \mathbf{u}^{\text{nom}}(\mathbf{x})\|_W^2, \\ \text{s.t.} \quad & \nabla_{\mathbf{x}} h^\top (f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} - \mathbf{v}_{\text{obs},l}(\tau)) \\ & + \gamma h(\mathbf{x}, \tau) \geq 0, \quad \forall \tau \in \mathcal{T}. \end{aligned} \quad (2)$$

Here $\|\mathbf{v}\|_W^2 = \mathbf{v}^\top W \mathbf{v}$ with $W \succ 0$, and $W = I$ in this paper. Eq. (2) is referred to as a semi-infinite program since, although the input consists of a finite number of components, the constraints extend over a continuous range. If a solution exists, there will only be one minimizer because the objective function is strictly convex and the feasible set is closed and convex within the compact set $\mathcal{U}$. In Sec. III, we replace the continuous constraints with a finite number of sufficient inequalities.

III. METHOD

In this paper, boundary geometry and semantic risk are fitted separately using a synchronized parameter, producing the $\mathbf{p}$ and s in **Assumption 1**. (Sec. III-A). We then check if the obstacle geometry is enclosed by the fitted curve (Sec. III-C). The controller uses the semantic risk values to calculate the semantics-aware safety distance, while we linearly transform the polynomial CBF constraint into a finite set of input-affine constraints in Bernstein coefficient representation (Sec. III-B). In Sec. III-C we make assumptions required to guarantee collision-avoidance.

A. Semantics-Aware Obstacle Representation

The safety distance requirements do not only depend on the Euclidean distance to the obstacle, but also on the object boundary where the hazard is located. Under the assumption of Sec. II-B, we define $\{\bar{\mathbf{p}}_k\}_{k=1}^{N_s} \subset \mathbb{R}^2$ to be the counter-clockwise ordered samples from a denoised point cloud or a known model geometry, and let $\{\bar{s}_k\}_{k=1}^{N_s} \subset [-1, 1]$ be their corresponding semantic labels. The label $\bar{s}_k$ is the risk value for a specific boundary location $\bar{\mathbf{p}}_k$, where -1 is the highest risk and 1 is the lowest. Ordered samples are placed on the boundary using the normalized parameter $\tau_k \in [0, 1]$, computed using the cumulative chord length:

$$\tau_k = \frac{\sum_{j < k} \|\bar{\mathbf{p}}_{j+1} - \bar{\mathbf{p}}_j\|}{\sum_{j \leq N_s} \|\bar{\mathbf{p}}_{j+1} - \bar{\mathbf{p}}_j\|}, \quad \tau_1 = 0, \quad \bar{\mathbf{p}}_{N_s+1} := \bar{\mathbf{p}}_1. \quad (3)$$

τ_k is the length of the sampled polygon from the first to the k th sample, divided by the total polygon perimeter. $\mathbb{S}^1$ is represented with $\tau = 0$ and $\tau = 1$ considered the same point. Since τ is shared by both geometry and risk, a risk value $s(\tau)$ is assigned to each boundary point $\mathbf{p}(\tau)$. Fig. 1(a) and (b)

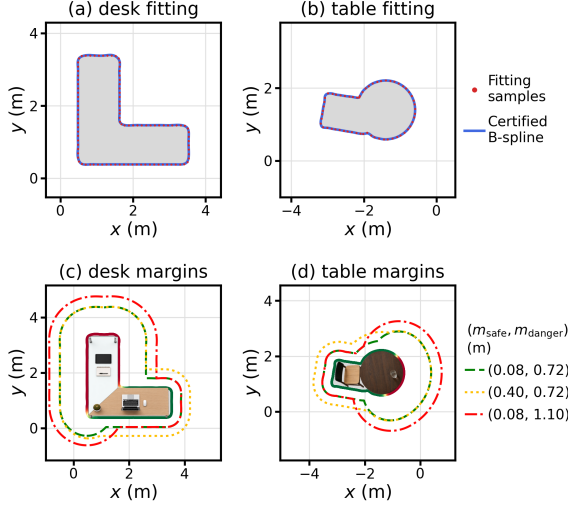


Fig. 1. B-spline boundaries and safety margins for an L-shaped desk and a table with a chair. In (a) and (b), gray represents the reference geometry, red dots are boundary samples that are shifted outward for fitting, and blue curves are the fitted B-splines that enclose the reference geometry. The outward shifts were $\rho = 0.016$ m in (a) and 0.031 m in (b). The mean gap between the L-shaped desk enclosure and the reference boundary is 0.017 m, with a maximum gap of 0.029 m. (c,d) Comparison of margins for enclosures from the household objects in Fig. 4, where $\rho = 0.016$ and 0.031 m for the L-shaped desk and the table/chair. Green colors represent safe regions and red represents hazardous regions along the boundary. Different curves represent the zero level of h_{sem} for each of the three margin pairs indicated in the lower legend.

demonstrates sampled points and the curve enclosures. Their semantics-aware margins are shown in Fig. 1(c) and (d).

We use periodic cubic B-splines with the same K basis functions $b_j(\tau)$ on $\tau \in [0, 1]$:

$$\mathbf{p}(\tau) = \sum_{j=0}^{K-1} b_j(\tau) \mathbf{P}_j, \quad s(\tau) = \sum_{j=0}^{K-1} b_j(\tau) S_j \quad (4)$$

to fit both geometry and risk. The coefficients of the B-spline are the geometric control points $\mathbf{P}_j \in \mathbb{R}^2$ for the B-spline of the fitted curve, and the control values $S_j \in \mathbb{R}$ for the B-spline of the semantic score. The basis weight $b_j(\tau)$ measures j 's contribution at a given τ . The curve consists of cubic polynomial segments connected at knots. The interval enclosed by two adjacent knots is called a knot span. The parameter interval is equally divided into K spans and each has a length of $1/K$. The periodic connection ensures that the first and second derivatives are continuous (C^2 continuity) at all knots, including where the start and end of the curve meet to fit the curve into an enclosed boundary.

Eq. (4) shows every weighted control point summed into a single curve together. A degree- d B-spline has at most $d+1$ nonzero basis functions in each span. Let $\tau \in [j/K, (j+1)/K]$ on span j and let us utilize the local parameter $\xi = K\tau - j \in [0, 1]$:

$$\begin{aligned} \mathbf{p}_j(\xi) &:= \mathbf{p}((j+\xi)/K) \\ &= N_0(\xi) \mathbf{P}_{j-1} + N_1(\xi) \mathbf{P}_j \\ &\quad + N_2(\xi) \mathbf{P}_{j+1} + N_3(\xi) \mathbf{P}_{j+2}. \end{aligned} \quad (5)$$

We periodically circulate the subscripts of the control points, so that $\mathbf{P}_{-1} = \mathbf{P}_{K-1}$, $\mathbf{P}_K = \mathbf{P}_0$, and $\mathbf{P}_{K+1} = \mathbf{P}_1$. The window including the four control points is shifted by one as we move to the next span. This results in the adjacent spans sharing three control points. The local basis functions N_i do not change and what we fit is the vector coefficients $\mathbf{P}_j$. Substituting $r = \xi+1, \xi, \xi-1, \xi-2$ into the centered cubic B-spline $\varphi(r)$ of Eq. (7), we get N_0, N_1, N_2, N_3 , respectively:

$$\begin{aligned} &[N_0(\xi), N_1(\xi), N_2(\xi), N_3(\xi)] \\ &= \frac{1}{6} [(1-\xi)^3, 3\xi^3 - 6\xi^2 + 4, -3\xi^3 + 3\xi^2 + 3\xi + 1, \xi^3]. \end{aligned} \quad (6)$$

The sum of these functions is 1, and the functions are not negative. Likewise, we get $s_j(\xi) := s((j+\xi)/K) = \sum_{i=0}^3 N_i(\xi) S_{j-1+i}$. C^2 connection is guaranteed in the geometry and risk even at the closing point if we use the shared controls with the basis.

The global basis of Eq. (4) is defined by the centered cubic B-spline $\varphi(r)$ [15]. The four cubic polynomial pieces are defined:

$$\varphi(r) = \frac{1}{6} \begin{cases} (r+2)^3, & -2 \leq r < -1, \\ 4 - 6r^2 - 3r^3, & -1 \leq r < 0, \\ 4 - 6r^2 + 3r^3, & 0 \leq r < 1, \\ (2-r)^3, & 1 \leq r < 2, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

$r \in \mathbb{R}$ is dimensionless and not normalized. $\varphi(r)$ is nonzero only when $|r| < 2$. $\xi+1, \xi, \xi-1, \xi-2$ traverse $[1, 2]$, $[0, 1]$, $[-1, 0]$, and $[-2, -1]$, respectively as ξ moves from 0 to 1. Thus, each individual N_i is a piece of φ on the local span.

First, we construct a periodic basis for the whole curve. Let k index the global basis functions:

$$b_k(\tau) = \sum_{n \in \mathbb{Z}} \varphi(K\tau - k - nK), \quad k = 0, \dots, K-1.$$

Here, n identifies copies of the same basis function shifted by integer periods in τ . Increasing τ by 1 only changes the numbering of these copies, so $b_k(\tau+1) = b_k(\tau)$.

Next, consider span j , where $\xi = K\tau - j$ and hence $K\tau = j + \xi$. Its four control points have indices $k = j-1+i$, $i = 0, \dots, 3$, interpreted periodically. Substituting this index into the argument of the associated basis function gives

$$K\tau - k = (j + \xi) - (j - 1 + i) = \xi + 1 - i.$$

So the four local weights become $N_i(\xi) = \varphi(\xi + 1 - i)$.

Since φ is 0 outside the support, only these four basis functions contribute to the span. Thus, the global sum in Eq. (4) becomes the local version expression in Eq. (5) represented with only four terms. $\mathbf{P} \in \mathbb{R}^{K \times 2}$ and $\bar{\mathbf{P}} \in \mathbb{R}^{N_s \times 2}$ include geometric control points and observations. With the design matrix $\Phi_{k,j} = b_j(\tau_k)$, we use ridge regression to get the control points:

$$\mathbf{P} = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top \bar{\mathbf{P}}, \quad \lambda = 10^{-4}. \quad (8)$$

The semantic fit utilizes the control values and labels $S \in \mathbb{R}^K$ and $\bar{s} \in \mathbb{R}^{N_s}$, with the design matrix and ridge weight

as above in addition to box constraints:

$$S = \arg \min_{-1 \leq S \leq 1} \|\Phi S - \bar{s}\|_2^2 + \lambda \|S\|_2^2, \quad (9)$$

which is a bounded linear least-squares problem.

Lemma 2 (Codomain from the control points). *Let $S_j \in [-1, 1]$ for each j . Since the periodic cubic B-spline basis is nonnegative and sums to 1 on $\mathbb{S}^1$ [15], $s(\tau)$ is a weighted sum of S_j . Thus, for all τ , $s(\tau) \in [-1, 1]$.*

Without the box constraints of Eq. (9), the fitted safety clearance can be lower than m_{safe} . S can be refitted to incorporate the semantic labels that can often change on household objects such as the ones in Fig. 1. For example, if the robot is carrying a water cup, the risk on the desk boundary near the laptop should be refitted to incorporate the semantic risk on the boundary around it.

Remark (Curve representation choice). *We constructed the obstacle boundary using piecewise cubic B-splines. The CBF polynomial degrees are at most six for each span. Bézier segments with transformed control points can be used over the shared parameter intervals with the B-spline. Additionally, one might alternatively use a single global Bézier curve to fit the boundary. In our comparison, the B-spline representation achieved higher accuracy in fitting the shape. Under the fair parameter settings for each approach, the outward offsets were adjusted until the enclosure test passed. The B-spline representation reduced the mean distance between the ground truth boundary and the fitted curve by 78.0% for the three shapes from Fig. 4 compared to the single global Bézier curve.*

B. Semantics-Aware Barrier and Coefficient Constraints

Given $(\mathbf{p}(\tau), s(\tau))$ and the disk robot with radius $R > 0$, we assign a safety distance to each point on the obstacle boundary. Each obstacle comes with its own parameter and this gives $\mathcal{T} = \bigsqcup_{l=1}^L \mathbb{S}^1$. On component l , $\mathbf{p}(\tau) = \mathbf{p}^l(\tau)$ and $s(\tau) = s^l(\tau)$. For $\mathbf{x} \in \mathbb{R}^2$ and $\tau \in \mathcal{T}$, the barrier function and semantics-aware margin are given by

$$h(\mathbf{x}, \tau) = \|\mathbf{x} - \mathbf{p}(\tau)\|^2 - [R + m(s(\tau))]^2, \quad (10)$$

$$m(s) = m_{\text{safe}} + \kappa(1 - s), \quad \kappa = (m_{\text{danger}} - m_{\text{safe}})/2, \quad (11)$$

where we omitted the obstacle index l of m for brevity. The margins $m_{\text{danger}} \geq m_{\text{safe}} \geq 0$ are user-defined, and $m(s)$ linearly interpolates along the clearance based on the risk. Lemma 2 gives $m(s(\tau)) \in [m_{\text{safe}}, m_{\text{danger}}]$. With $C(\tau) = R + m(s(\tau)) > 0$, we can define the minimum barrier and its 0-superlevel safe set as

$$h_{\text{sem}}(\mathbf{x}) = \min_{\tau \in \mathcal{T}} h(\mathbf{x}, \tau), \quad \mathcal{H}_{\text{sem}} = \{\mathbf{x} \mid h_{\text{sem}}(\mathbf{x}) \geq 0\}. \quad (12)$$

Remark. *The barrier function in Eq. (12) takes the minimum over a continuous family of smooth squared barriers. While it is locally Lipschitz, it may not be smooth, so Lemma 1 does not apply directly. Sec. III-C instead uses a comparison argument for each smooth barrier. The final feedback controller is assumed to be locally Lipschitz within its operating*

domain. If the feedback is discontinuous, a different analysis is needed, such as using Filippov solutions [16].

$\mathbf{x} \in \mathcal{H}_{\text{sem}}$ if and only if $\|\mathbf{x} - \mathbf{p}(\tau)\| \geq R + m(s(\tau))$ for all $\tau \in \mathcal{T}$. So each boundary point keeps the robot center at least $C(\tau)$ away from it. Note the distances are unsigned so points inside the boundary may be in the defined safe set. Thus, we must place the robot in the set region outside of each fitted curve of the objects. Readers may refer to Corollary 1 for detailed discussion. For reporting and planning purposes, we use clearance in meters, $c_{\text{sem}}(\mathbf{x}) = \min_{\tau \in \mathcal{T}} [\|\mathbf{x} - \mathbf{p}(\tau)\| - R - m(s(\tau))]$. The individual barriers $h(\cdot, \tau)$ are smooth and $\nabla_{\mathbf{x}} h = 2(\mathbf{x} - \mathbf{p}(\tau))$.

There is no need to differentiate or evaluate the minimum in Eq. (12) to calculate the control input, as we apply a CBF inequality constraint for each smooth $h(\mathbf{x}, \tau)$.

1) *A polynomial constraint on each original span:* For a fixed boundary parameter τ , the semantic label fixed on the object boundary remains the same over time. If we differentiate h with respect to time, we can obtain the condition $\dot{h} + \gamma h = \Psi \geq 0$, where

$$\Psi(\mathbf{x}, \mathbf{v}, \tau) = 2(\mathbf{x} - \mathbf{p}(\tau))^\top \mathbf{v} + \gamma h(\mathbf{x}, \tau). \quad (13)$$

$\mathbf{v} = \mathbf{v}_{\text{rel}} = \mathbf{f} + g\mathbf{u} - \mathbf{v}_{\text{obs}}(\tau)$ is the robot's relative velocity with respect to the boundary point. $\mathbf{v}_{\text{obs}} = 0$ for static obstacles, and equations for moving boundaries are defined in Sec. III-C. We limit τ to each span j using $\tau = (j + \xi)/K$ at a fixed state. $\mathbf{v}_{\text{rel},j}$ is the relative velocity here. For brevity, we are omitting state arguments. Let

$$\begin{aligned} \mathbf{w}_j &= \mathbf{x} - \mathbf{p}_j(\xi), & D_j &= \mathbf{w}_j^\top \mathbf{w}_j, \\ A_j &= \mathbf{w}_j^\top \mathbf{v}_{\text{rel},j}, & C_j &= R + m(s_j(\xi)). \end{aligned} \quad (14)$$

Then

$$\Psi_j(\xi, \mathbf{u}) = 2A_j(\xi, \mathbf{u}) + \gamma[D_j(\xi) - C_j(\xi)^2] \quad (15)$$

is at most degree 6 in ξ and affine in $\mathbf{u}$. $\mathbf{w}_j$ and C_j are at most degree 3 in ξ . Relative velocity is constant in ξ under translation and at most degree 3 in ξ under rigid-body rotation.

We can obtain a sufficient condition for the entire span if we expand in the 6th degree Bernstein basis:

$$\begin{aligned} \Psi_j(\xi, \mathbf{u}) &= \sum_{k=0}^6 \beta_{jk}(\mathbf{x}, \mathbf{u}) B_k^6(\xi), \\ B_k^6(\xi) &= \binom{6}{k} \xi^k (1 - \xi)^{6-k}. \end{aligned} \quad (16)$$

First, we expand Eq. (15) in powers: $\Psi_j = \sum_{r=0}^6 q_{jr} \xi^r$. $q_{jr}(\mathbf{x}, \mathbf{u})$ is affine in control input, and the coefficients for powers that are not used are zero. The basis transformation follows:

$$\beta_{jk} = \sum_{r=0}^k \frac{\binom{k}{r}}{\binom{6}{r}} q_{jr} = \mathbf{a}_{jk}(\mathbf{x})^\top \mathbf{u} + c_{jk}(\mathbf{x}). \quad (17)$$

Each span provides 7 affine coefficient constraints, $\beta_{jk} \geq 0$.

Theorem 1 (Nonnegativity condition of Bernstein coefficients). *If, for every original knot span j of every obstacle, all the coefficients satisfy $\beta_{jk}(\mathbf{x}, \mathbf{u}) \geq 0$, $k = 0, \dots, 6$, then the CBF condition $\Psi(\mathbf{x}, f + g\mathbf{u} - \mathbf{v}_{\text{obs}}(\tau), \tau) \geq 0$ holds for all $\tau \in \mathcal{T}$.*

Proof. At any $\xi \in [0, 1]$, the sum of the nonnegative Bernstein weights is 1, so $\Psi_j(\xi, \mathbf{u}) \geq \min_k \beta_{jk}(\mathbf{x}, \mathbf{u}) \geq 0$. The original knot spans cover the entire $\mathcal{T}$, so the inequalities are satisfied on all boundary parameters. $\square$

The coefficient test is conservative because a polynomial can remain nonnegative even when some of its Bernstein coefficients are negative [17]. It might thus reject an input that satisfies the continuum condition, or it might be infeasible even though the continuum problem has a feasible input.

2) *One finite optimization:* With the Euclidean input constraint used in the experiments, the optimization becomes:

$$\begin{aligned} \mathbf{u}^*(\mathbf{x}) = \arg \min_{\mathbf{u} \in \mathcal{U}} \quad & \frac{1}{2} \|\mathbf{u} - \mathbf{u}^{\text{nom}}(\mathbf{x})\|_W^2 \\ \text{s.t.} \quad & \mathbf{a}_{jk}(\mathbf{x})^\top \mathbf{u} + c_{jk}(\mathbf{x}) \geq 0, \\ & \|\mathbf{u}\|_2 \leq u_{\max}, \end{aligned} \quad (18)$$

for $k = 0, \dots, 6$ on every span j of every obstacle. The input set $\mathcal{U}$ contains the magnitude limit $u_{\max} > 0$. For the unicycle, we add the actuator limits in Sec. IV-B as well. This is a convex Quadratically Constrained Quadratic Program (QCQP) with a unique minimizer whenever feasible. Before adding the input constraints, there are $7 \sum_l K_l$ barrier inequalities, giving 896 for two obstacles with $K_l = 64$. On each update, coefficient constraints are assembled, and the nominal input is used unchanged as long as it passes the coefficient and input limit constraints. If it does not, then the convex program is solved. Theorem 1 assumes exact arithmetic. In our implementation, we check every returned coefficient and input limit, and reject any output from the solver that does not pass these tests.

C. Theoretical Analysis

1) *Enclosure:* Let $\mathbf{p} : \mathbb{S}^1 \rightarrow \mathbb{R}^2$ be a closed fitted curve for a compact and connected obstacle $\mathcal{O}_l \subset \mathbb{R}^2$. To include non-simple curves, we define the enclosed region as follows:

$$\Omega(\mathbf{p}) = \mathbf{p}(\mathbb{S}^1) \cup \{\mathbf{q} \in \mathbb{R}^2 \setminus \mathbf{p}(\mathbb{S}^1) \mid \text{wind}(\mathbf{p}, \mathbf{q}) \neq 0\}, \quad (19)$$

where $\mathbf{p}(\mathbb{S}^1)$ is the curve and $\text{wind}(\mathbf{p}, \mathbf{q})$ is its winding number around a point $\mathbf{q}$ not on the curve, counting counterclockwise turns as positive and clockwise turns as negative. For a simple counterclockwise curve, $\text{wind}(\mathbf{p}, \mathbf{q})$ is 1 inside and 0 outside.

Prior to fitting with Eq. (8), the points are translated in the direction of their outward unit normals: $\bar{\mathbf{p}}_k \leftarrow \bar{\mathbf{p}}_k + \rho \mathbf{n}_k$, $\rho > 0$. We repeat the fitting process until the resulting curve passes the enclosure test. Starting from this passing value, we decrease the uniform offset until the test first fails. We refine this pass/fail transition to 10^{-4} m and keep an accepted ρ . This procedure relies only on an enclosure certificate and does not depend on finding the smallest possible offset.

For the enclosure check, we represent each cubic B-spline span in Bernstein form, i.e., as a cubic Bézier segment with identical shape on $\xi \in [0, 1]$,

$$\mathbf{p}_j(\xi) = \sum_{i=0}^3 B_i(\xi) \mathbf{B}_{j,i}, \quad B_i(\xi) = \binom{3}{i} \xi^i (1 - \xi)^{3-i}. \quad (20)$$

Here, B_i are scalar basis functions and $\mathbf{B}_{j,i}$ are the Bézier control points. The transformation is given exactly by

$$\begin{bmatrix} \mathbf{B}_{j,0} \\ \mathbf{B}_{j,1} \\ \mathbf{B}_{j,2} \\ \mathbf{B}_{j,3} \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{P}_{j-1} \\ \mathbf{P}_j \\ \mathbf{P}_{j+1} \\ \mathbf{P}_{j+2} \end{bmatrix}. \quad (21)$$

Because the nonnegative Bernstein weights sum to unity, each segment must lie inside the convex hull of its control points: $\mathbf{p}_j([0, 1]) \subseteq \text{conv}\{\mathbf{B}_{j,i}\}$. We apply De Casteljau subdivision [15] repeatedly to obtain smaller subarcs with tighter convex hulls, which we then test against the obstacle.

Lemma 3 (Sufficient conditions for obstacle enclosure). *Given a finite subdivision $\{I_j\}$ of the knot spans, let $\mathcal{C}(I_j)$ be the convex hull of the Bézier control points of $\mathbf{p}(I_j)$. Assume that $\mathcal{C}(I_j) \cap \mathcal{O}_l = \emptyset$ for every j , and $\text{wind}(\mathbf{p}, \mathbf{q}_0) \neq 0$ for some $\mathbf{q}_0 \in \mathcal{O}_l$. Then $\mathcal{O}_l \subseteq \text{int } \Omega(\mathbf{p})$.*

Proof. The hull condition and $\mathbf{p}(I_j) \subseteq \mathcal{C}(I_j)$ ensure that the entire curve is disjoint from $\mathcal{O}_l$. Because winding is locally constant away from the curve itself and because $\mathcal{O}_l$ is connected, the winding number must equal $\text{wind}(\mathbf{p}, \mathbf{q}_0) \neq 0$ for all points of $\mathcal{O}_l$. Each such point must therefore have an open neighborhood around it with nonzero winding. By Eq. (19), this neighborhood lies inside $\Omega(\mathbf{p})$ and so the point is in the interior of $\Omega(\mathbf{p})$. $\square$

We start by testing each of the original knot spans. If a span has a convex hull disjoint from the obstacle, it is accepted. If a span has a convex hull that intersects the obstacle, we bisect the interval and test both halves. If we reach the maximum subdivision depth with an intersection, the certificate fails. Otherwise, the accepted subintervals form $\{I_j\}$.

To determine the winding, we construct a polygon by connecting the endpoints of the accepted intervals. Within its convex hull, each subarc is homotopic to the straight line between its endpoints. Because this convex hull is free of obstacles, this homotopy preserves the winding number, so the winding number of the polygon is the winding number of the curve.

Corollary 1 (Collision avoidance from certified enclosure). *Under the hypotheses of Lemma 3, consider a robot disk $\mathcal{R}(\mathbf{x}) = \{\mathbf{r} \in \mathbb{R}^2 : \|\mathbf{r} - \mathbf{x}\| \leq R\}$ with $R > 0$ and a continuous center trajectory satisfying $\text{wind}(\mathbf{p}, \mathbf{x}(0)) = 0$ and $h_{\text{sem}}(\mathbf{x}(t)) \geq 0$. Then, throughout the trajectory, $\text{dist}(\mathbf{x}(t), \mathcal{O}_l) \geq R + m_{\text{safe}}$ and $\mathcal{R}(\mathbf{x}(t)) \cap \mathcal{O}_l = \emptyset$.*

Proof. Since the score-range lemma (Lemma 2) implies $m \geq m_{\text{safe}} \geq 0$, non-negativity of the minimum semantic

barrier implies $\|\mathbf{x} - \mathbf{p}(\tau)\| \geq R > 0$ for all τ , so that the continuous center trajectory avoids the curve and so preserves its initial winding number of zero. Any point $\mathbf{q} \in \mathcal{O}_l$ in the true obstacle has non-zero winding by the enclosure lemma (Lemma 3), so that the line segment joining the current center to that point must intersect the curve at some $\mathbf{c} = \mathbf{p}(\tau^*)$, and neither endpoint lies on the curve, so that the intersection point must lie strictly between them. Thus, $\|\mathbf{x} - \mathbf{q}\| = \|\mathbf{x} - \mathbf{c}\| + \|\mathbf{c} - \mathbf{q}\| > \|\mathbf{x} - \mathbf{c}\| \geq R + m(s(\tau^*)) \geq R + m_{\text{safe}}$. $\square$

2) *Moving boundaries and ordinary closed-loop solutions*: We model each dynamic obstacle as a rigid body. The fitted curve $\hat{\mathbf{p}}^l$ and the true obstacle $\hat{\mathcal{O}}_l$ go through the same continuously differentiable translation $\mathbf{c}_l(t)$ and planar rotation $\mathbf{Q}_l(t) \in \text{SO}(2)$. Thus, their positions in the world frame are $\mathbf{p}_t^l(\tau) = \mathbf{c}_l(t) + \mathbf{Q}_l(t)\hat{\mathbf{p}}^l(\tau)$ and $\mathcal{O}_l(t) = \mathbf{c}_l(t) + \mathbf{Q}_l(t)\hat{\mathcal{O}}_l$, respectively. t denotes time here and to consider this motion, we need to include time in the state $\mathbf{z} = (\mathbf{x}, t)$, with $\dot{t} = 1$. The semantic labels are fixed in the body frame of the obstacles, so the required clearance $C(\tau)$ does not change with respect to time for a fixed τ . Thus we get $\partial_t C = 0$. The barrier is $h(\mathbf{z}, \tau) = \|\mathbf{x} - \mathbf{p}_t^l(\tau)\|^2 - C(\tau)^2$. The boundary point's velocity is $\mathbf{v}_{\text{obs},l}(\tau) = \dot{\mathbf{c}}_l + \dot{\mathbf{Q}}_l \hat{\mathbf{p}}^l(\tau)$, which is of maximum degree 3 in τ on each original spline span. The barrier's gradient with respect to the augmented state is:

$$\nabla_{\mathbf{z}} h(\mathbf{z}, \tau) = (2(\mathbf{x} - \mathbf{p}_t^l(\tau)), -2(\mathbf{x} - \mathbf{p}_t^l(\tau))^\top \mathbf{v}_{\text{obs},l}(\tau)). \quad (22)$$

The CBF condition is

$$\Psi_{\mathbf{z}}(\mathbf{z}, \mathbf{u}, \tau) = 2(\mathbf{x} - \mathbf{p}_t^l(\tau))^\top (f + g\mathbf{u} - \mathbf{v}_{\text{obs},l}(\tau)) + \gamma h(\mathbf{z}, \tau). \quad (23)$$

With the substitution $\mathbf{v} = \mathbf{v}_{\text{rel}}$ into Eq. (13) we have this condition, so the Bernstein theorem (Theorem 1) applies at each fixed augmented state $\mathbf{z}$. Time is omitted for static obstacles. Other robot states are still present in the augmented state, but their zero gradient entries are omitted from the above gradient formula Eq. (22). Clearly the Bernstein-feasible set is a subset of the continuum-feasible set:

$$\mathcal{B}_{\text{sem}}(\mathbf{z}) = \{\mathbf{u} \in \mathcal{U} : \Psi_{\mathbf{z}}(\mathbf{z}, \mathbf{u}, \tau) \geq 0 \ \forall \tau \in \mathcal{T}\}, \quad (24)$$

$$\mathcal{B}_{\text{B}}(\mathbf{z}) = \{\mathbf{u} \in \mathcal{U} : \beta_{jk}(\mathbf{z}, \mathbf{u}) \geq 0 \ \forall j, k\} \subseteq \mathcal{B}_{\text{sem}}(\mathbf{z}).$$

Theorem 2 (Conditional semantics-aware forward invariance). *Let the dynamics on an open operating domain $\mathcal{N}$ of the augmented state $\mathbf{z}$ be control affine and locally Lipschitz. At every state $\mathbf{z} \in \mathcal{N}$ assume that the final feedback selects an input $\mathbf{k}(\mathbf{z}) \in \mathcal{B}_{\text{B}}(\mathbf{z})$. Also assume that this feedback $\mathbf{k} : \mathcal{N} \rightarrow \mathcal{U}$ is locally Lipschitz. Then, for any initial state $\mathbf{z}(0) \in \mathcal{N}$ whose minimum semantic barrier $h_{\text{sem}}(\mathbf{z}(0))$ is nonnegative, the unique closed-loop solution satisfies*

$$h_{\text{sem}}(\mathbf{z}(t)) \geq e^{-\gamma t} h_{\text{sem}}(\mathbf{z}(0)) \geq 0 \quad (25)$$

as long as the solution exists within the operating domain $\mathcal{N}$.

Proof. Because the dynamics and feedback are locally Lipschitz, a unique local closed-loop solution $\mathbf{z}(t)$ is guaranteed. Along this solution, $\mathbf{k}(\mathbf{z}(t))$ satisfies the Bernstein constraints. The Bernstein theorem (Theorem 1), along with the chain rule, then gives the following condition for every fixed $\tau \in \mathcal{T}$:

$$\frac{d}{dt} h(\mathbf{z}(t), \tau) + \gamma h(\mathbf{z}(t), \tau) \geq 0.$$

We get $\frac{d}{dt} [e^{\gamma t} h(\mathbf{z}(t), \tau)] \geq 0$ by multiplying by $e^{\gamma t}$, and $h(\mathbf{z}(t), \tau) \geq e^{-\gamma t} h(\mathbf{z}(0), \tau)$ by integrating over $[0, t]$. This holds as long as $\mathbf{z}(t)$ stays in $\mathcal{N}$. Since the boundary parameter τ was arbitrary, taking the minima over the compact set of boundary parameters $\mathcal{T}$ proves the bound in Eq. (25) for the minimum semantic barrier. $\square$

For the controller in Eq. (18) to satisfy Theorem 2, the optimization must remain feasible throughout the operating domain $\mathcal{N}$, and its minimizer must be locally Lipschitz within the operating domain. These conditions are assumed in the continuous-time guarantee. Readers may refer to [18] for detailed discussion on feasibility conditions for CBF-based optimization problems. Conditions where such optimization-based controllers are continuous are discussed in [19].

Suppose each obstacle is enclosed according to Lemma 3, and $\text{wind}(\mathbf{p}_t^l, \mathbf{x}(0)) = 0$. Under the hypotheses of Theorem 2, the minimum semantic barrier stays nonnegative. Now applying Corollary 1, while the solution exists in $\mathcal{N}$, $\mathcal{R}(\mathbf{x}(t)) \cap \bigcup_l \mathcal{O}_l(t) = \emptyset$ is guaranteed. The same applies to obstacle body frames since enclosure and winding do not change under rigid motion.

IV. RESULTS

We evaluate the experiments with moving obstacles and navigation in a household environment in a simulated environment. The units for lengths and times are meters and seconds, respectively, and γ is in seconds⁻¹. We use $m_{\text{safe}} = 0.08$ and $\Delta t = 0.02$ for our B-spline approach. In the pedestrian example we used $R = 0.36$, $\gamma = 2.5$, $\|\mathbf{u}\| \leq u_{\text{max}} = 1.5$, $K = 64$. The disk based approach used $r_{\text{safe},l} = r_{\text{enc},l} + R + m_{\text{safe}}$. We used 2.17 m and 1.62 m for the radii for the cart carrier and the box carrier to be tightly enclosed.

A. Crossing Pedestrians

In Fig. 2, the single-integrator robot moves from $(4.0, -2.0)$ to $(-4.0, 2.0)$. The cart and box carriers move at 0.65 m/s while one follows $x = -1.29$ and the other follows $x = 0.84$. Their m_{danger} values are 1.35 and 0.80, respectively, and this allows the hazardous region to be formed in front of the pedestrians as the robot should not attempt to awkwardly fit in front of the pedestrians carrying objects.

We compare our approach with the disk based $h_{\text{base}}(\mathbf{x}) = \|\mathbf{x} - \mathbf{c}_l\|^2 - r_{\text{safe},l}^2$ from [20]. Our method increases the travel distance and time by 0.49 m and 4.60 s. However, the proportions of the path where the mean semantic score satisfies $\bar{s} < 0$ and $\bar{s} < -0.3$ decreased from 61.8%

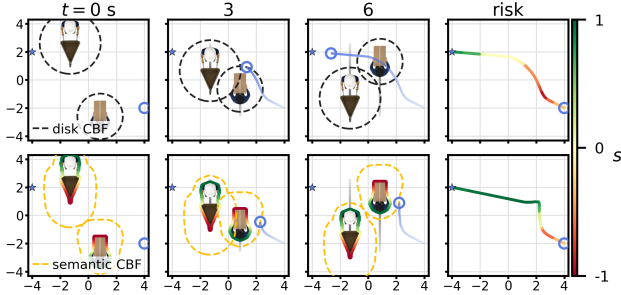


Fig. 2. The last column colors the robot’s path based on the risk at the nearest fitted boundary point. The human images show the fixed shapes that are used to model each person. Red and green represent hazardous and safe fitted boundaries, the blue circle is the robot, and the blue star is the goal. The dashed curves show the inflated disk safety boundary in the top row and the semantic safety boundary in the bottom row. The baseline method (top) passes in front of the box, while our method (bottom) passes behind both people. The minimum robot-center distances to the sampled fitted box/cart boundaries are 0.43/0.57 m for the baseline and 0.54/2.72 m for our method. Minima are $c_{\text{disk}} = \pm 0.0018$ m and $c_{\text{sem}} = \pm 0.0160$ m²

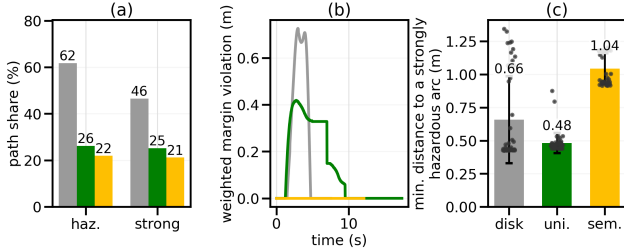


Fig. 3. Risk in Fig. 2, with gray for the disk baseline, green for uniform spline margins, and yellow for ours. (a) Path share (%) where the mean semantic score $s < 0$ (haz.) or $s < -0.3$ (strong). (b) Weighted margin violation $\sum_l \max\{0, -s_l\} \max\{0, R + m(s_l) - d_l\}$ (m) versus time (s), where d_l and s_l are the center distance and score at the nearest fitted point of pedestrian l . It integrates to 1.645, 2.237, and 0 m s for the disk, uniform, and semantic methods, respectively, at the evaluated states. The semantic controller yields zero because it enforces the clearances used in this metric. (c) Minimum robot-center distance (m) to either sampled fitted strongly hazardous arc over the same 50 pairs completed by all three controllers (seed 3), shown as mean $\pm$ population SD and individual dots. Metrics use the original semantic labels, 1500 boundary points per obstacle, and states every 20 ms.

to 21.9% and from 46.5% to 21.2%, respectively. When uniform margins ($\kappa = 0$) were used, these proportions were 26.2% and 25.2%, respectively, with a path length of 14.56 m. Our approach increased the minimum distance from the robot center to the strongly hazardous boundary arc from 0.44 m with uniform margins to 0.92 m.

In separate experiments, label noise with standard deviation $\sigma_s = 0.8$ and mislabeling 20% of the samples increased the mean path length from 10.02 m to 12.23 m and 11.97 m respectively. Geometric noise with $\sigma_g = 0.08$ m increased the maximum offset ρ from 0.016 m to 0.293 m. All 40 runs with such noise successfully reached the goal.

B. Evaluation with different semantic risk assignments

The robot in the household environment follows $\dot{\mathbf{x}} = (v \cos \theta, v \sin \theta)$ and $\dot{\theta} = \omega$, with $v_{\max} = 1$, $\omega_{\max} = 2.5$, and $m_{\text{danger}} = 0.72$. We hold the velocity $\mathbf{u}_k$ of the look-ahead point $\mathbf{y} = \mathbf{x} + \ell \mathbf{d}$ between updates, and here $\ell = 0.08$, $\mathbf{d} = (\cos \theta, \sin \theta)$, and $\mathbf{d}^\perp = (-\sin \theta, \cos \theta)$. The commands

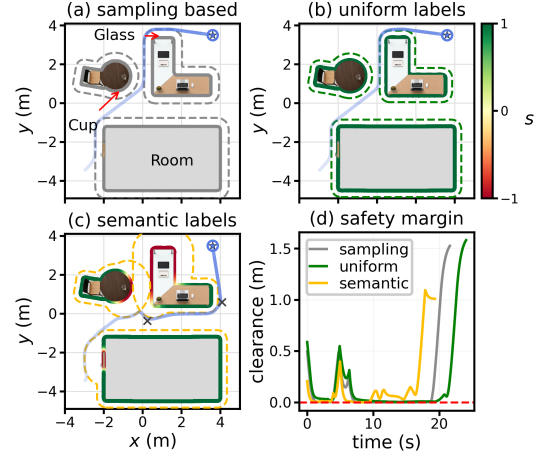


Fig. 4. This is a household comparison with (a) coverage-corrected sampling, (b) uniform margins, and (c) semantic margins. The green/red represents safe/hazardous boundaries, the dashed curves represent the zero levels, and the blue curves represent the robot paths. The crosses in (c) show the two temporary waypoints used to help the robot recover when it gets stuck. Panel (d) shows the robot’s clearance over time, using the sampled safety bound for (a) and c_{sem} for (b) and (c).

$v = \mathbf{d}^\top \mathbf{u}_k$ and $\omega = (\mathbf{d}^\perp)^\top \mathbf{u}_k / \ell$ get recalculated based on the current heading. In each update, the forward motion constraints are $\|\mathbf{u}_k\| \leq v_{\max}$, $\mathbf{d}^\top \mathbf{u}_k \geq 0$, and $|(\mathbf{d}^\perp)^\top \mathbf{u}_k| \leq \ell \omega_{\max}$. When the body radius is $r_{\text{body}} = 0.18$, the disk centered at $\mathbf{y}$ and whose radius is $R_\ell = r_{\text{body}} + \ell = 0.26$ includes the robot body.

The sampling based approach from Fig. 4 is from [10]. We used 16 samples on the circle that encloses the robot, and on each obstacle we used 200 uniform arclength samples. The clearance was increased by the sum of the sample covering radii. We apply the constraints on all robot–obstacle sample pairs that give the minimum values. All methods use the same nominal gain, physical input limits, and Clarabel solver.

All methods attempt to reach the goal using a nominal controller with a gain of 1.5, maintaining the commands inside the input limits. Upon deadlock, we used simplified A* waypoints and the same move to goal controller to reach each waypoint.

All three runs reach the goal. The sampling based and the uniform margin runs were successful without the need for a simplified planner. This resulted in paths of 11.75 m and 11.69 m, respectively. If we use the semantics-aware margin, the path between the table and the L-shaped desk is blocked, so two intermediate waypoints were generated by a planner and a path of 12.67 m was used. The uniform margin and the semantic margin filters rectified 74.8% and 64.6% of the nominal inputs, and the minimum sampled c_{sem} values were 0.0033 m and 0.0010 m, respectively. Compared to the uniform margin, the semantics-aware margin increased the minimum distance from the robot center to the reference watercup-side arc from 0.61 m to 0.87 m and added 0.98 m to the total travel distance. The clearance was checked using 2000 sampled boundary parameters per obstacle at every control step.

C. Constraint Conservatism

We evaluated how much conservatism Bernstein coefficient constraints impose on the control inputs in Fig. 2. We used 401 boundary parameters per span to approximate the input set that satisfies the continuum constraints. Across all 605 states, Bernstein coefficient constraints reduced the estimated input area by at most 0.09% on average and 0.23% at any state.

D. Quantitative Results

We compare the disk, uniform, and semantic-based controllers in Fig. 2's pedestrian crossing scenario. The start and goal positions are uniformly sampled with $x \in [3.5, 4.5]$, $y \in [-3, 1]$ for the starts and $x \in [-4.5, -3.5]$, $y \in [-1, 3]$ for the goals. We report results for 50 geometrically eligible start-goal paired runs completed by all three controllers. The uniform and semantic controllers share the fitted geometry and control settings. In the order of disk/uniform/semantic, the minimum distances from the robot center to the strongly hazardous arc were 0.66 ± 0.33 , 0.48 ± 0.08 , and 1.04 ± 0.12 m (Fig. 3(c)). The path lengths were 9.40 ± 1.02 , 11.12 ± 2.50 , and 9.24 ± 0.76 m (mean $\pm$ population SD). The mean exposure to hazard was 50%, 40%, and 39%, and the mean integrated weighted margin violation was 1.36, 1.41, and 0 m.s. All metrics use the same semantic labels for all three controllers.

Across five runs of the pedestrian crossing experiment on an AMD Ryzen 9 3900X desktop, the construction of the constraints and input computation took a median 2.78 ms per step, and every update was computed within the 20 ms control interval.

V. CONCLUSION

This paper presents a safe control framework that combines obstacle geometry with local semantic risk using smooth B-spline boundary representations and Bernstein coefficient constraints. The framework captures how risk varies along irregular obstacle boundaries and uses this information to determine how much clearance the robot should maintain in different regions. We tested the framework in simulations with crossing pedestrians and cluttered household environments. The robot maintains larger distances near hazardous regions, such as the front of pedestrians carrying objects, while preserving space near relatively safe regions compared with uniformly applying the largest margin. Local semantic information helps determine which route to take and how much clearance to maintain. As a result, our approach links the context of the real-world environment with low-level control, allowing robots to navigate intelligently in context-rich and dense environments.

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